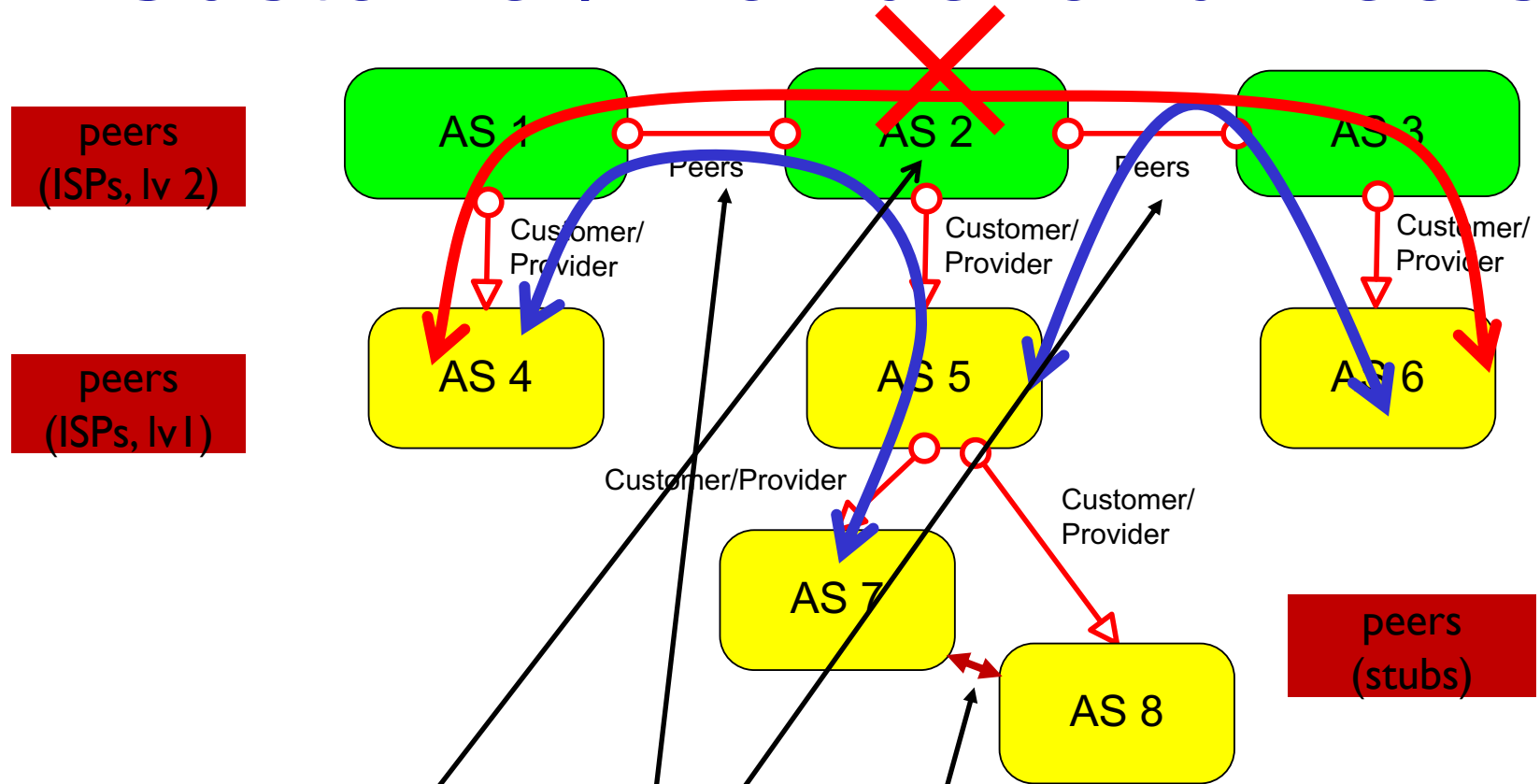
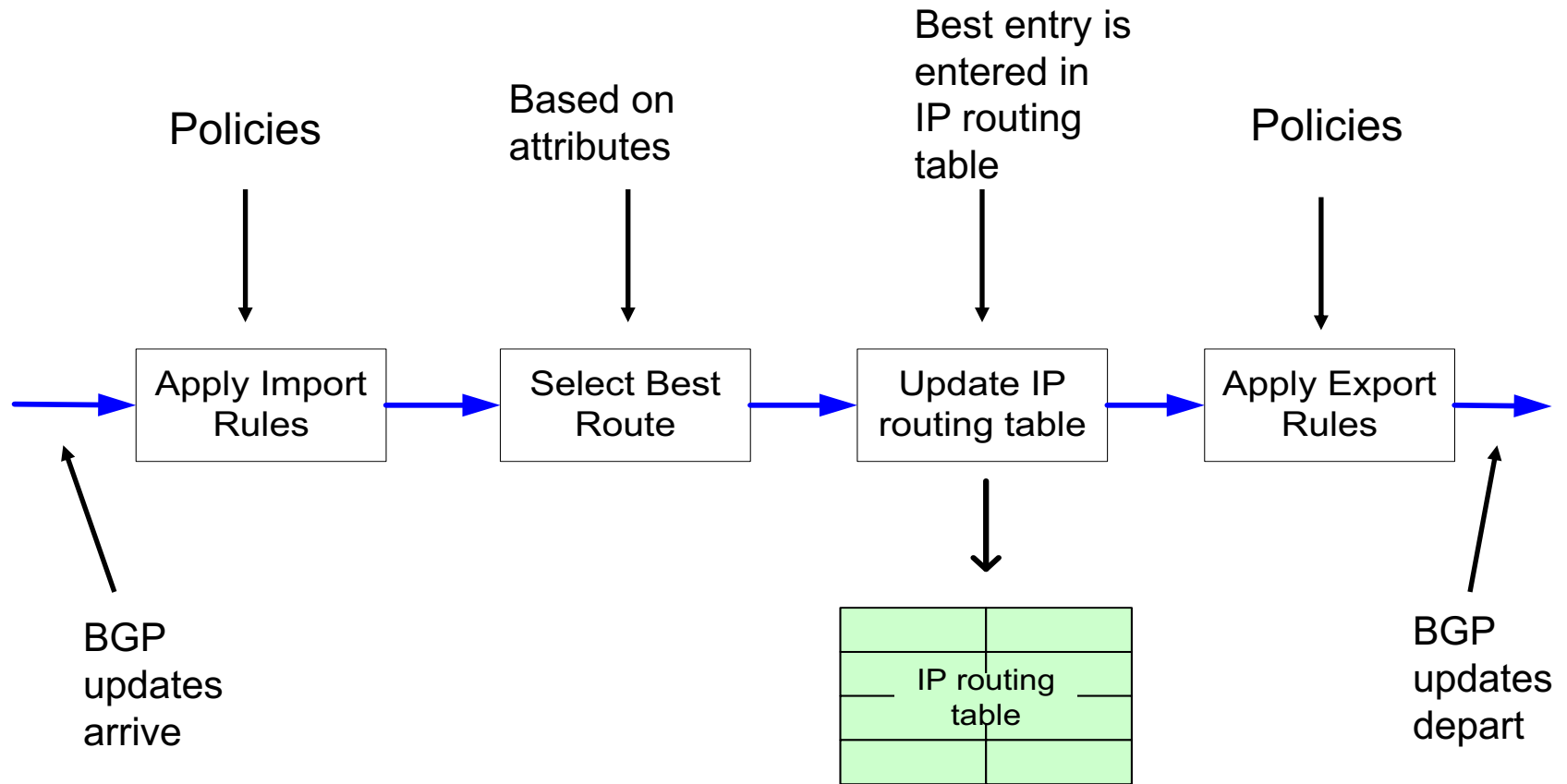


Customer/Provider and Peers



- **stubs** can have peer relationships – direct link, carries no transit
- **transit** networks can have a **peer** relationship
- **peers** provide **transit** between their **respective customers**
- **peers** do not provide **transit** between **peers**, i.e., traffic from AS1 to AS3 cannot go through AS2.
- **peers** have to go up one layer to reach another peer if not directly connected
- **peers** normally do not pay each other for service

Import and Export Policies



Why different Intra-, Inter-AS routing ?

policy:

- inter-AS: manager of an AS wants control over how its traffic is routed externally, and who routes through its net (not applicable for STUB networks).
- intra-AS: single admin, so no policy decisions needed

scale:

- hierarchical routing saves table size, reduced update traffic

performance:

- intra-AS: can **focus** on performance (e.g., cost)
- inter-AS: policy may **dominate** over performance

BGP interactions

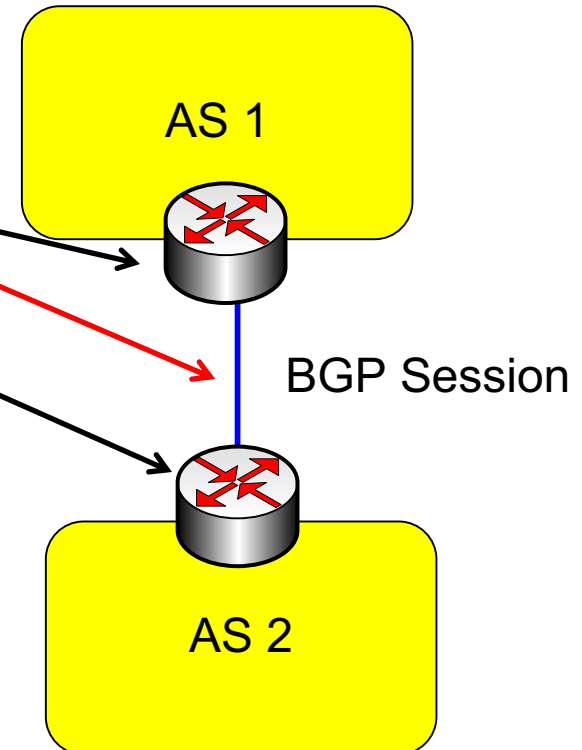
- BGP is executed between two routers

- BGP session
- BGP peers

- procedure:

1. establishes *TCP connection (port 175) to BGP peer*
2. exchange all BGP routes
3. as long as connection is alive:
Periodically send incremental updates

- **Note:** Not all autonomous systems need to run BGP. On many **stub** networks, the route to the provider can be **statically configured**



BGP messages

- BGP messages exchanged between peers over TCP session
- BGP messages:
 - **OPEN:** opens TCP connection to remote BGP peer (port 179) and authenticates sending BGP peer
 - **UPDATE:** advertises new path (or withdraws old)
 - **KEEPALIVE:** keeps connection alive in absence of UPDATES; also ACKs OPEN request
 - **NOTIFICATION:** reports errors in previous msg; also used to close connection

BGP message header



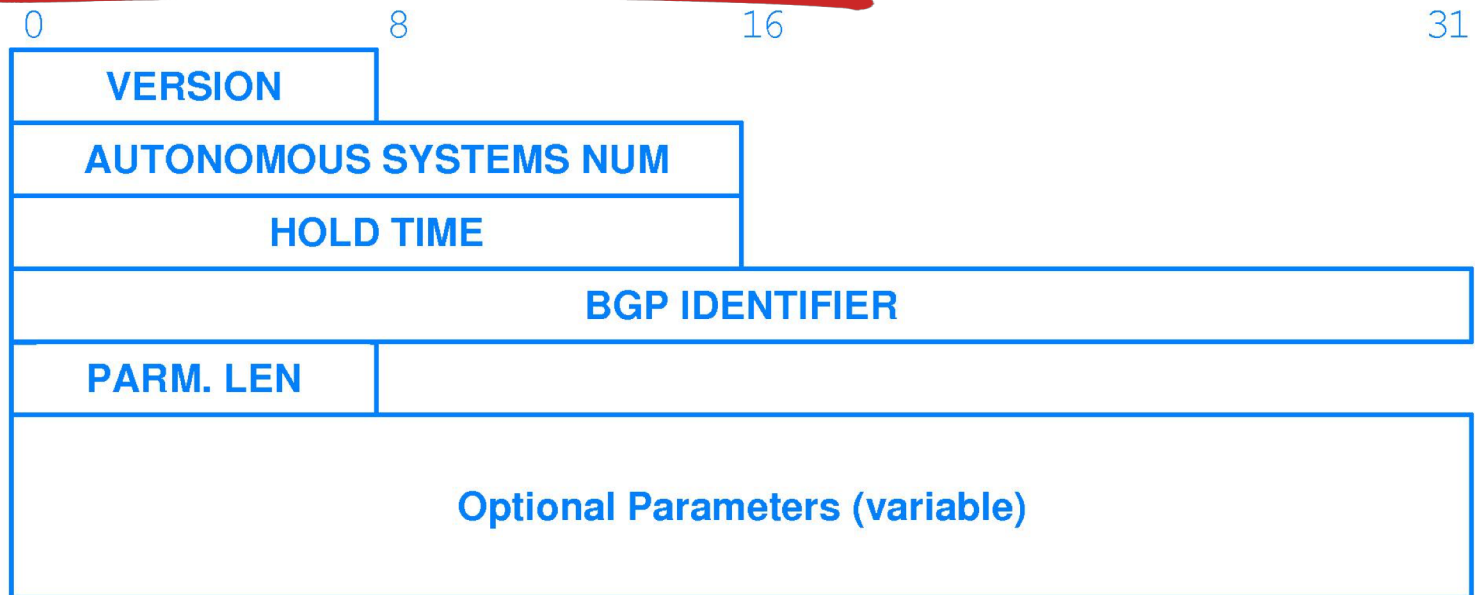
The format of the header that precedes every BGP message.

Marker is an agreed upon value (synchronization pattern) between two peers. Usually all one's, but can be used for authentication. Used to synchronize the two ends.

Length gives total message length in octets

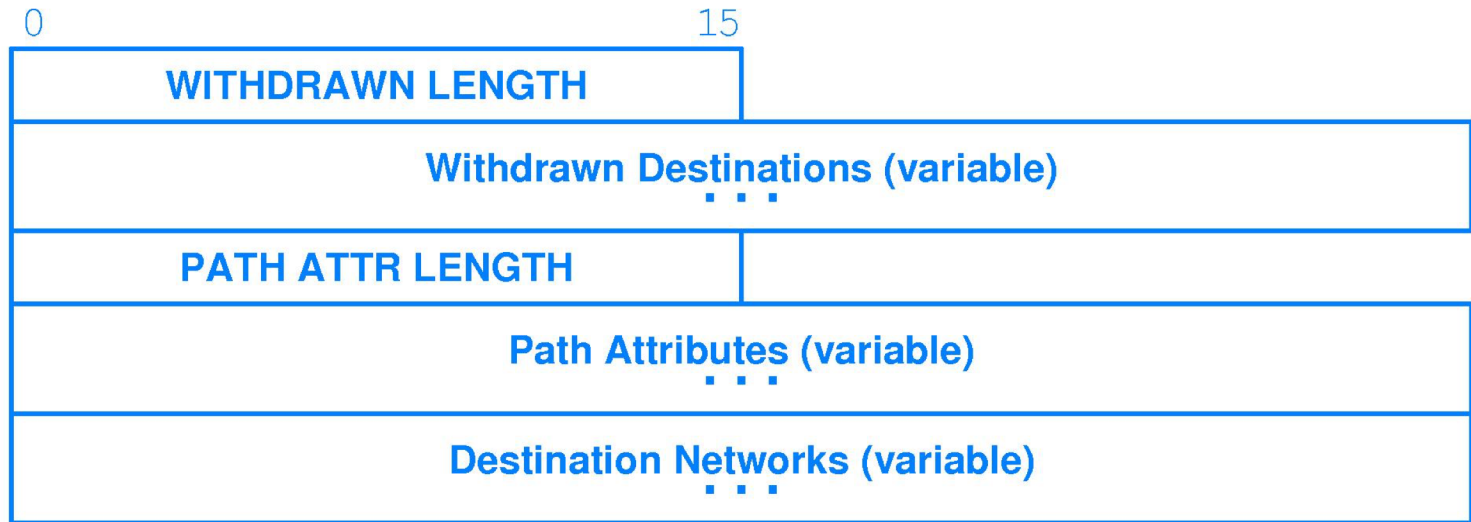
Type contains one of the message types shown in previous slide

BGP Open message



The format of the IPv4 BGP OPEN message that is sent at start-up. Octets shown in the figure follow the standard message header.

BGP Update message



BGP UPDATE message format in which variable size areas of the message may be omitted. These octets follow the standard message header.

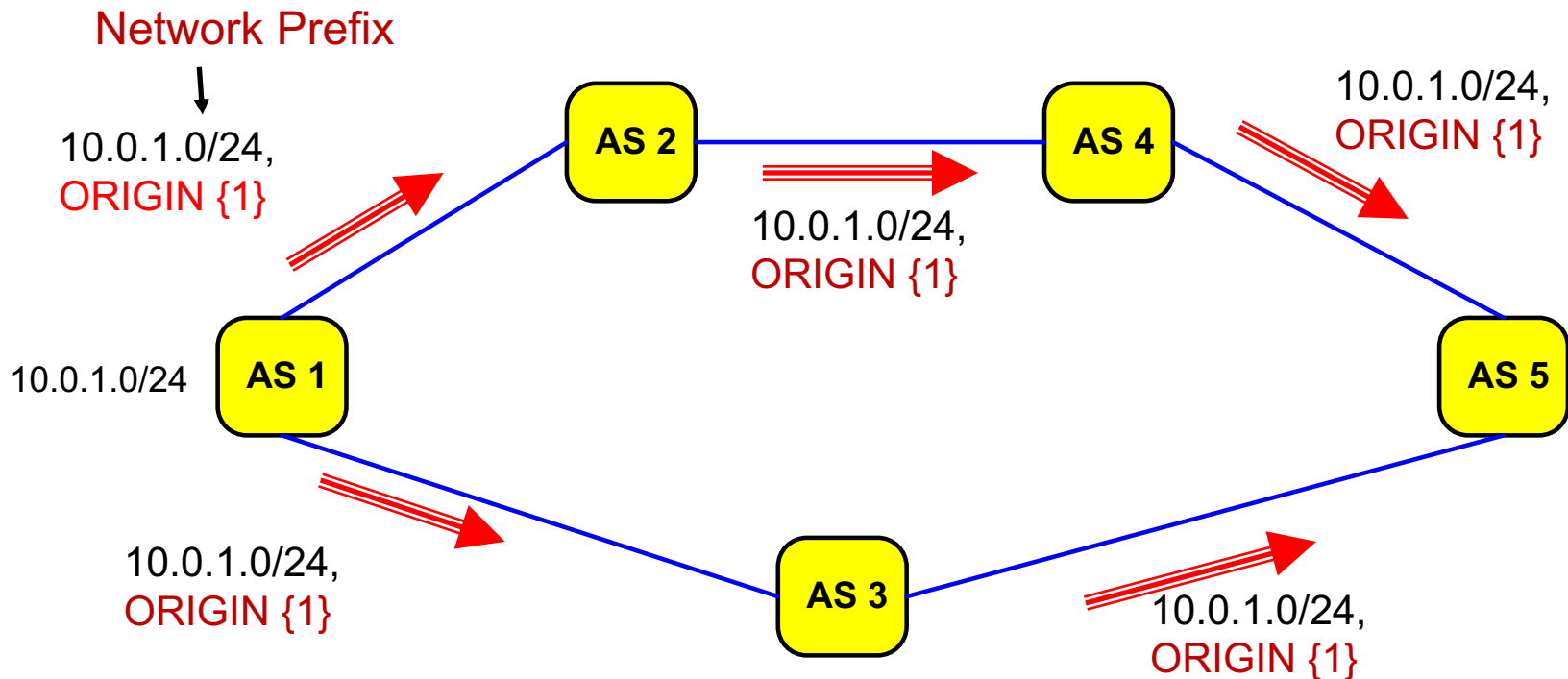
Note that any field labeled “**variable**”, can be omitted if there is no information for a parameter

BGP route updates

- BGP **route advertisement** is sent in a BGP UPDATE message
- a route is announced as a **Network Prefix**, e.g., 10.0.1.0/24, and **Attributes**
- **Attributes** specify details about a route:
 - Mandatory attributes:
 - ORIGIN**
 - AS_PATH**
 - NEXT_HOP**
 - many other attributes

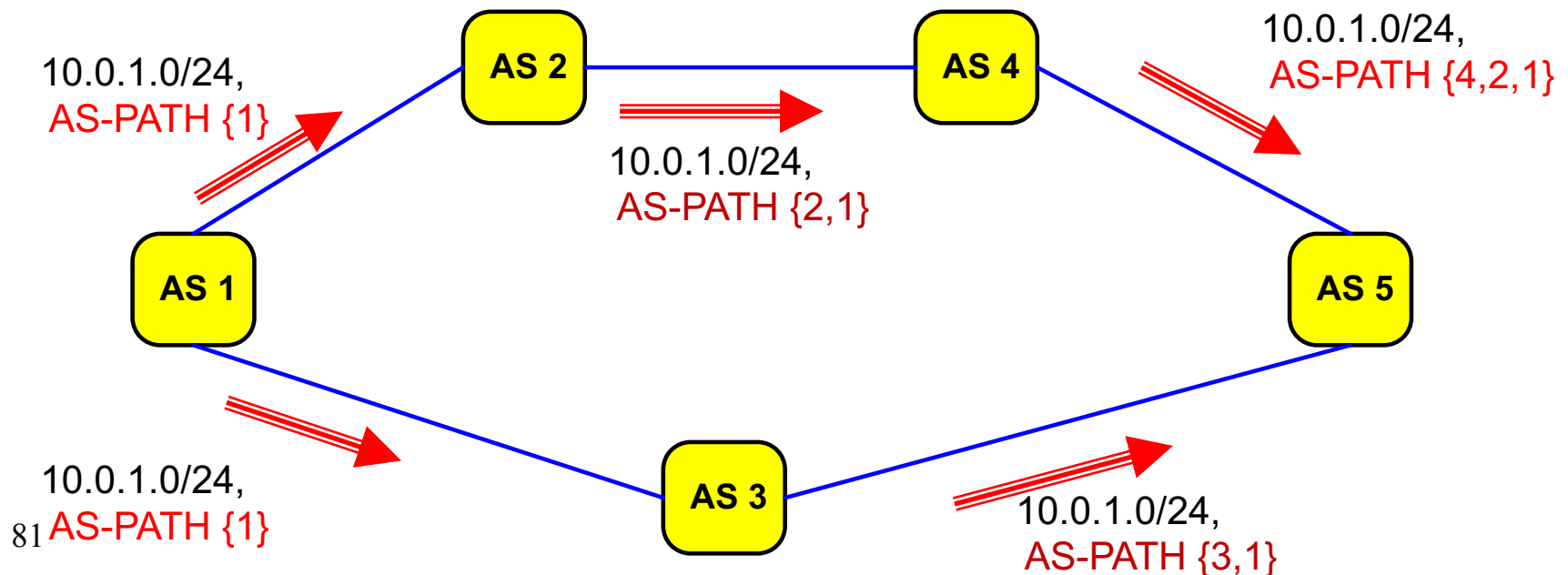
ORIGIN attribute

- originating domain sends a route to a network (here 10.0.1.0/24) with **ORIGIN** attribute (**AS number**)



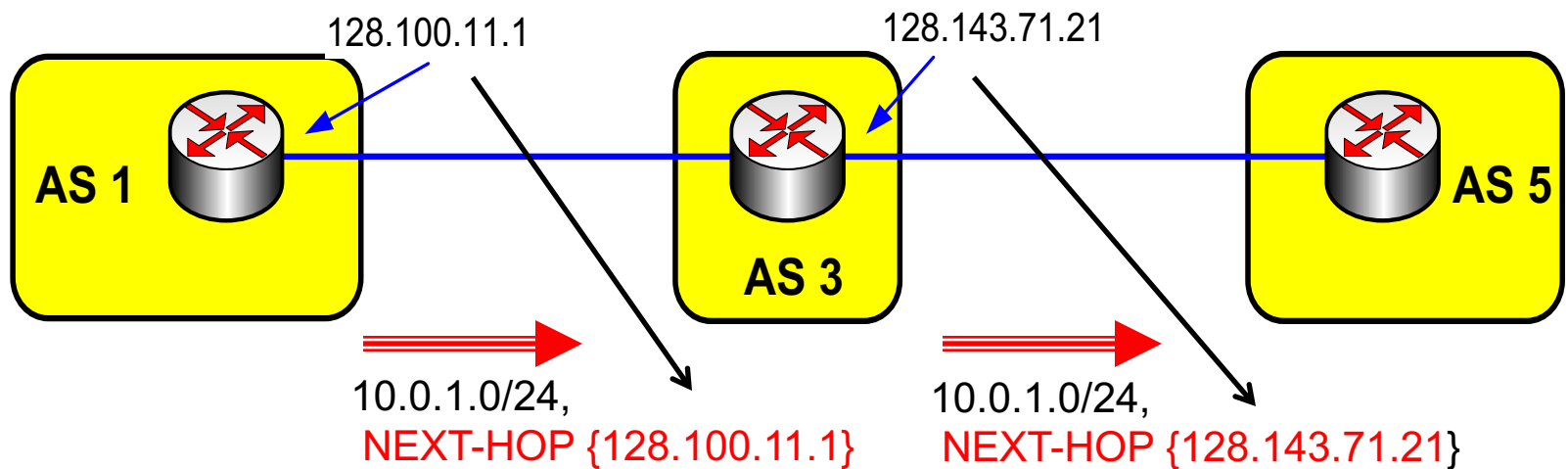
AS-PATH attributes

- each AS that propagates a route **prepends** its own **AS number**
 - AS-PATH creates a full path to reach the network prefix 10.0.1.0/24
- path information prevents routing loops from occurring
- path information also provides information on the length of a path (no. of ASes enroute, by default, a shorter route is preferred)
- **Note:** BGP aggregates routes according to CIDR rules



NEXT-HOP attributes

- each router that sends a route advertisement, includes its own IP address of the forwarding port in a **NEXT-HOP attribute**
- the attribute provides information for the routing table of the receiving router in the next AS on the path



BGP NEXT-HOP -> IGP information

